



THE BIG BANG THEORY

THE BIG BANG THEORY

Before reading:

- What do you already know about the Big Bang Theory?



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Before the Big Bang

The Big Bang was the origin event of the universe. About 14 billion years ago, there was only energy. The energy was incredibly hot and dense.

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Before the Big Bang

Before the Big Bang, time did not exist, matter did not exist, space did not exist. Time, matter, and space all came into existence with the Big Bang. Suddenly, the universe expanded and cooled which allowed for the energy to convert into matter and antimatter.





The expanding universe cooled from its incredibly hot beginning. It was calculated to have started with a temperature of 10^{32}°C , that's 100 000 000 000 000 000 000 000 000 000 $^{\circ}\text{C}$!

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Timeline of the Big Bang

Within about 1 second of its existence, the universe cooled down enough for **protons**, **neutrons**, and **electrons** to form.



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proton: the positively charged subatomic particle in the nucleus of the atom

neutron: the neutral subatomic particle in the nucleus of the atom

electron: the negatively charged subatomic particle that orbits the nucleus of the atom

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Timeline of the Big Bang

After about 3 minutes, the temperature of the universe cooled below one billion degrees Celsius (1 000 000 000°C). At this “cooler” temperature, protons and neutrons formed the **nuclei** of **hydrogen** and **helium**.



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nuclei: the plural of nucleus

hydrogen: the smallest and simplest element

helium: the second smallest and simplest element, after hydrogen

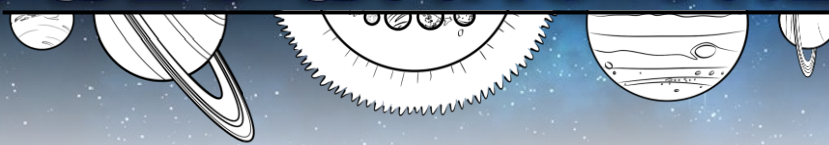
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Timeline of the Big Bang

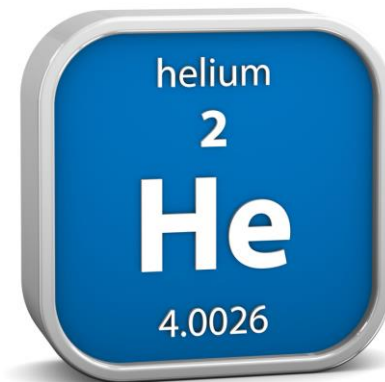
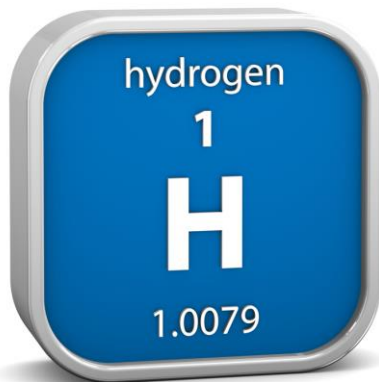
The universe continued to expand and cool, and by the time it was 300 000 years old its temperature was about 3000°C – cool enough for the hydrogen and helium nuclei to capture electrons and form atoms.

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Timeline of the Big Bang

After 100 million years the atoms of gas came together due to gravity and the first stars formed. The earliest stars were made of mostly hydrogen with some helium.

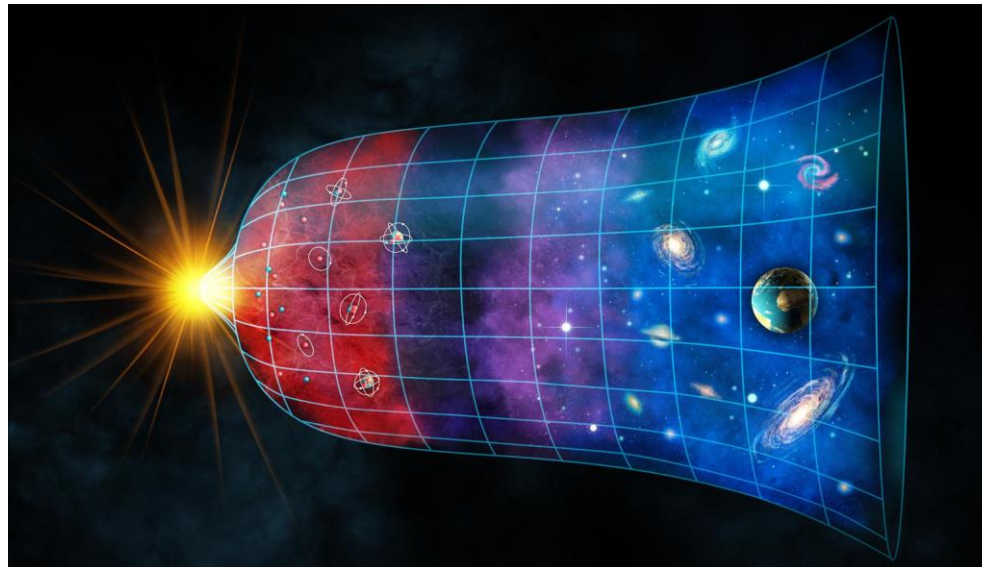


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Timeline of the Big Bang

Gravity was able to play a bigger role as the universe cooled, making a more structured universe with stars and galaxies.



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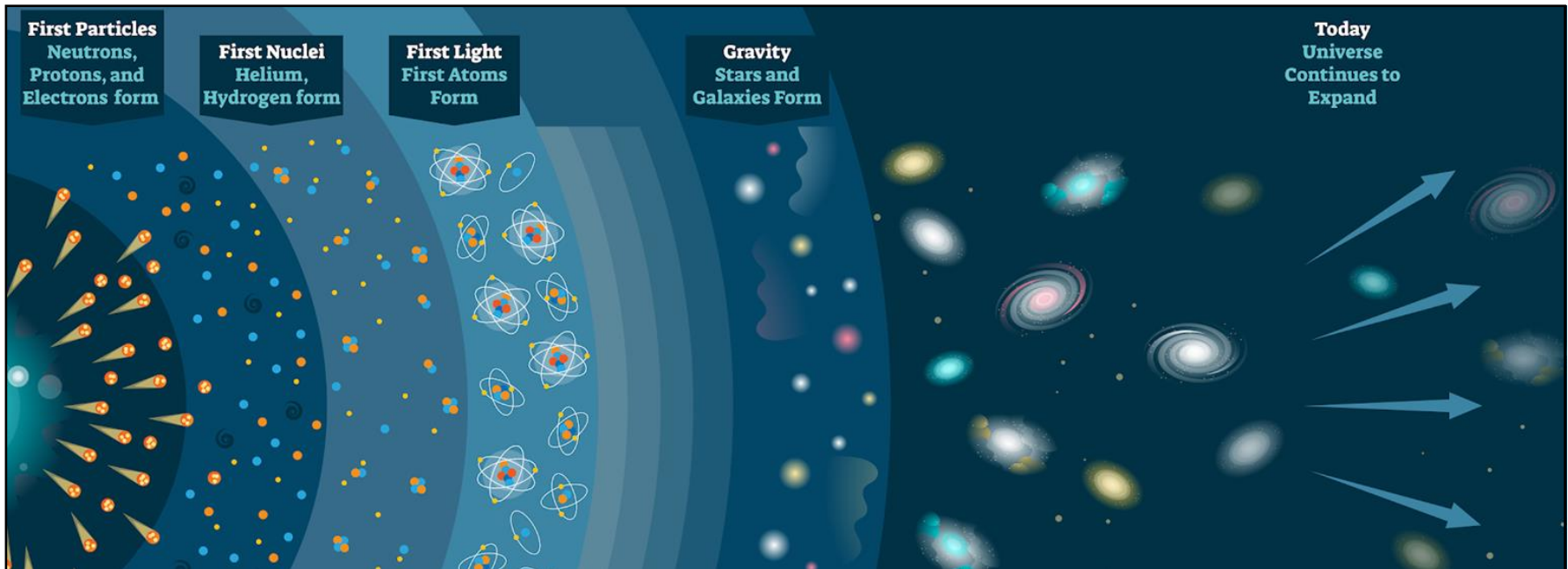


Discover the Diagram:

- 1. Read the labels on the diagram.
- 2. Explain what you are learning.
- 3. How does this diagram help you learn what you read?

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Timeline of the Big Bang



As time went on the universe grew and its average temperature decreased allowing gravity to pull together matter to form stars and galaxies.

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Evidence Supporting the Big Bang

In 1927, the Belgian priest and astronomer Georges Lemaître proposed the Big Bang **theory**.



theory: a well-constructed explanation that uses facts and observations



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Evidence Supporting the Big Bang

How did he theorize that the universe began with the Big Bang? The same way any scientist makes claims – he looked at the data.

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Evidence Supporting the Big Bang
Cosmologists used multiple sources of **observational evidence** to support the Big Bang theory,

cosmologist: an astronomer that studies the universe's origin

What do you think “observational evidence” means?



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Evidence Supporting the Big Bang

These are called spectral lines. The frequencies of the spectral lines are unique to each element – like a fingerprint.

Hydrogen



Helium



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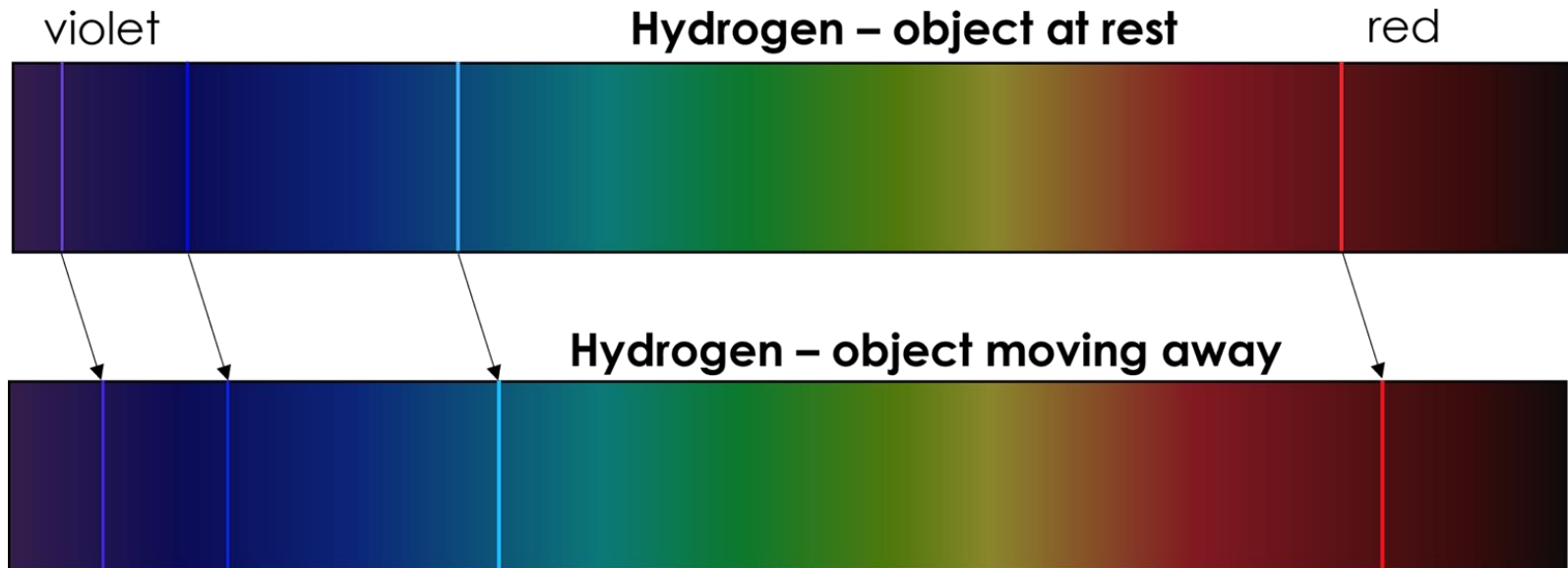


Evidence Supporting the Big Bang

Astronomers observed the light from distant galaxies as having these spectral lines for gases like hydrogen and helium, but the observed frequencies were lower than the known frequencies.

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Evidence Supporting the Big Bang



Hydrogen's spectral lines are its identifying "fingerprints". When the light source is moving away, its spectral lines are shifted toward the red end of the spectrum.

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The lowering of the frequencies of the spectral lines is called **red shifting**.

redshift: the stretching of light from a source moving away from the observer, resulting in the spectral lines shifted toward the red end of the spectrum



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Evidence Supporting the Big Bang

It is just like the Doppler Effect, when you hear a lower frequency sound when a fire truck drives away from you, except for light waves instead of sound waves.



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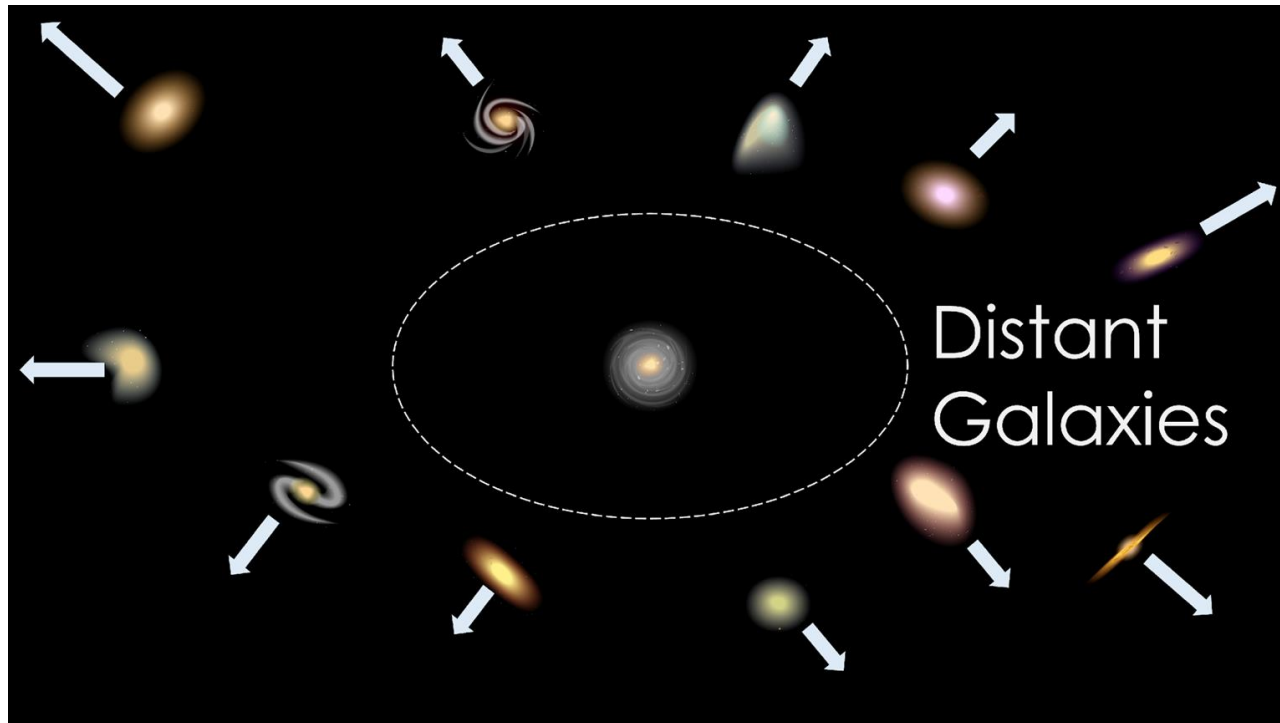
Evidence Supporting the Big Bang

In every direction astronomers look in distant space they see the galaxies' light is redshifted, which means *all* distant galaxies are moving away from us. Astronomers even calculated the speed of the galaxies using the Doppler Effect equation.

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Evidence Supporting the Big Bang

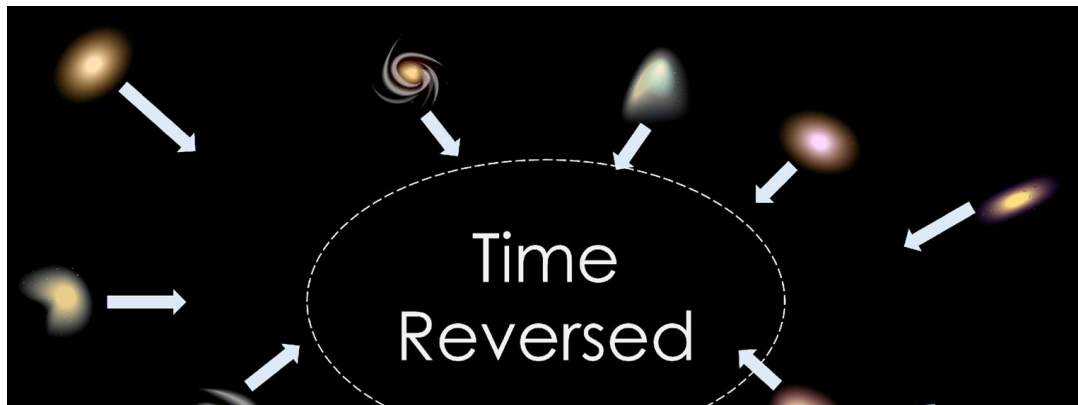
All distant galaxies are currently moving away from us.



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Evidence Supporting the Big Bang

Imagine time could rewind. If time was reversed, the distant galaxies would all move toward one another and would **converge** to a single point.



converge: to come together



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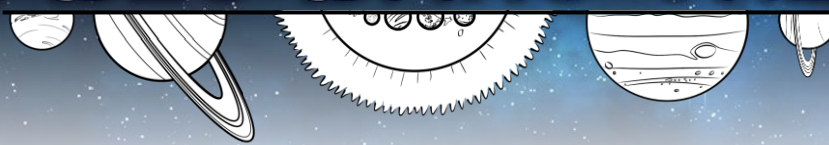


Evidence Supporting the Big Bang

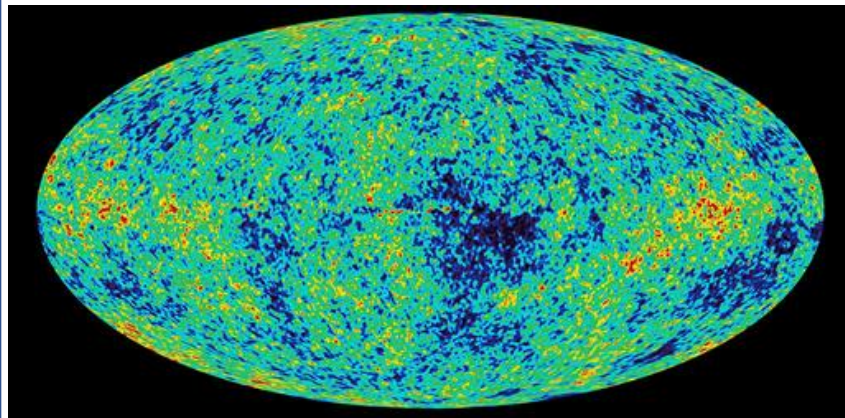
Redshifted light is not the only line of evidence supporting the Big Bang Theory. Observations of the **cosmic microwave background** and the relative abundance of elements like hydrogen and helium among other evidence supports the Big Bang Theory.



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cosmic microwave background: radiation from the early universe that has stretched to lower-energy radiation as the universe expanded, it is observed in every direction



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Evidence Supporting the Big Bang

If the observational evidence does not agree with the theory, that theory fails. The Big Bang theory continues to prevail and agrees with our wealth of observational data.

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Q1.

What did you learn that was new to you? Start with, “Wow, I never knew...”

What new information do you feel like you missed? If you are having trouble recalling, reread to clarify.

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Q2.

Imagine you were Georges Lemaître. Summarize your proof of the Big Bang theory.



ANY QUESTIONS?

